Android Ethical Hacking Tools Explained

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What apps usually ethical hackers use in android device?

Ethical hackers (penetration testers and security researchers) use Android devices for two main purposes: **testing mobile apps directly on the phone** or **turning the phone into a portable network audit tool**.

The exact tools depend on whether the Android device is **rooted** (unlocked for deep system access) or **non-rooted**.

1. Full Mobile Security Environments

Instead of installing single apps, security professionals often run dedicated penetration testing distributions on top of Android.

Kali NetHunter: The gold standard for mobile pentesting. It runs a full Kali Linux environment inside Android. It gives access to wireless injection, HID keyboard attacks (BadUSB), and hundreds of pre-installed security tools. *(Requires a rooted device and custom kernel for full hardware access).* www.kali.org

Termux: An advanced terminal emulator and Linux environment that runs without needing root. Hackers use Termux to run Python, Node.js, Metasploit, Nmap, and command-line network tools directly on the phone.

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These tools are used to scan networks, map connected devices, and inspect network traffic.

Nmap (via Termux): The most popular network discovery and port-scanning tool. It identifies active devices, open ports, and operating systems on a local Wi-Fi network.

Fing: A polished network scanner used to identify all connected hardware on a Wi-Fi network, detect hidden cameras, and run port scans.

Wireshark / tshark (or PCAP Remote): Used to capture and analyze network packets in real time to inspect unencrypted traffic. EC-Council

Aircrack-ng / Wifite (via NetHunter): Used for Wi-Fi security auditing, testing WPA/WPA2 password strength, and capturing handshakes. EC-Council

3. Web & API Traffic Interception

To test the security of mobile applications, ethical hackers intercept the traffic traveling between the Android app and the backend server. Bugcrowd

Burp Suite / OWASP ZAP (Proxy Setup): While the main application runs on a PC, hackers route their Android device's Wi-Fi proxy through Burp Suite or OWASP ZAP to inspect and modify HTTP/HTTPS requests sent by mobile apps.

HTTP CanHttpRequest / Packet Capture (Android Apps): Apps like *HttpCanary* or *Reqable* allow developers and security researchers to inspect SSL/TLS traffic directly on the phone without an external PC.

4. Reverse Engineering & App Analysis

Ethical hackers analyze Android application packages (`.apk` files) to check for hardcoded API keys, sensitive information, or logic flaws. Bugcrowd

JADX / Jadx-GUI: Decompiles Android DEX and APK files into readable Java source code to inspect how the application functions internally.

Frida & Objection: Dynamic instrumentation toolkits. Hackers inject scripts into running apps to bypass security controls like SSL pinning (which prevents traffic inspection) or root detection. Bugcrowd + 1

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MobSF (Mobile Security Framework): An automated framework used for static and dynamic vulnerability analysis of Android applications. Bugcrowd

5. OSINT & Reconnaissance

Over-The-Air or open-source intelligence tools used to collect data:

Shodan: Used to search for connected devices, exposed webcams, servers, and open ports around the world.

Termux Scripts: Tools like `theHarvester` or custom Python scripts are routinely executed inside Termux to gather domain and email intelligence.

Important Context: *Many of these security apps are not hosted on the standard Google Play Store due to platform policies. Researchers typically source them via open-source repositories like* **GitHub** *or the* **NetHunter Store.** *Using these tools without prior authorization on networks or systems you do not own is illegal.* EC-Council

Kali NetHunter

Kali NetHunter is the official mobile penetration testing platform developed by OffSec (Offensive Security) for Android devices. It turns an Android phone into a full-featured portable security auditing tool capable of executing advanced network, wireless, and physical attacks. Wikipedia + 1

1. Core Editions

NetHunter is offered in three primary editions, depending on your device's level of access:

Feature	NetHunter Rootless	NetHunter Lite	NetHunter
Device Requirement	Unmodified, Stock Android	Rooted (Magisk / KernelSU)	Rooted + C Kernel

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Feature	NetHunter Rootless	NetHunter Lite	NetHunter
Command Line & Tools	Yes (via Termux/PRoot)	Yes (full chroot)	Yes (full ch
GUI Desktop (KeX)	Yes	Yes	Yes
Wi-Fi Frame Injection	No	No	**Yes**
HID (BadUSB) Attacks	No	No	**Yes**
Bluetooth Arsenal	No	No	**Yes**

2. Key Features & Attack Modules

NetHunter KeX (Kali Desktop Experience): Allows you to mirror a full Kali Linux XFCE desktop directly on the phone or output it to an external monitor via HDMI or wireless display. Kali Linux

HID Keyboard Attacks (BadUSB): Emulates a USB keyboard when plugged into a target PC. It can automatically execute malicious scripts (Rubber Ducky style) at high speed.

BadUSB MITM Attacks: Turns the phone into an emulated USB network adapter, forcing target computers to route all network traffic through the Android phone for interception.

MANA / Wifipumpkin3 Evil Twin Attacks: Deploys rogue access points to mimic trusted Wi-Fi networks, capturing credentials through custom captive portals.

Wireless Frame Injection: Using supported external (or built-in) Wi-Fi chips, it allows deauthentication attacks, packet crafting, and WPA/WPA2 handshake sniffing.

Bluetooth Arsenal: Tools for scanning, spoofing, jamming, or delivering audio injections to nearby Bluetooth devices. Kali Linux

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1. **The Chroot/Container:** NetHunter installs a Linux *chroot* environment (or container) inside Android. This gives access to the entire Kali APT package repository without overwriting the phone's main Android OS.
2. **The NetHunter App:** An Android application that acts as a control center, providing a graphical menu to launch complex terminal commands, services, and attack scripts.
3. **Custom Kernels:** For full functionality (like packet injection and HID emulate drivers), NetHunter replaces the stock Android Linux kernel with a purpose-built kernel compiled with extra network and hardware drivers. Kali Linux

Want to learn how to set up NetHunter Rootless without modifying your phone?

Where can i download it?

Where you download **Kali NetHunter** depends on whether you want to install the **full system** or just the **NetHunter Android App Store** to get specific tools.

1. Official Full Images & Kernels

To download the full NetHunter OS image for your phone or tablet (Full, Lite, or Rootless):

Official Website: kali.org/get-kali/#kali-mobile

Direct Image Server: kali.download/nethunter-images/

Tip: *On the main page, scroll down to the* **NetHunter** *section to find pre-built zip files for supported devices (OnePlus, Samsung, Nexus, Xiaomi, etc.).*

2. NetHunter App Store (APK)

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If you only want to download specific security apps, the terminal client, or the **KeX**

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Web Store: store.nethunter.com

App Client APK: You can download the **NetHunter Store App** directly from store.nethunter.com or from their official GitLab Repository.

3. NetHunter Rootless (Via Termux)

If you do not want to root your phone, you don't download a traditional `.zip` image. Instead:

1. Download the **Termux** app from the **NetHunter Store** or F-Droid (do not use the Google Play version, as it is outdated).
2. Run the official installation script directly from OffSec inside Termux:

```bash
termux-setup-storage
pkg update && pkg install wget -y
wget -O install-nethunter-termux https://offs.ec/2MceZWr
chmod +x install-nethunter-termux
./install-nethunter-termux
```

Where would you like to go next?

Step-by-step guide for NetHunter Rootless ↪

How to flash NetHunter on a rooted phone ↪

I want one with all, most of features

To get a **Kali NetHunter setup with ALL (or most) features**—like Wi-Fi packet injection, HID keyboard attacks (BadUSB), Bluetooth sniping, and Evil Twin APs—you cannot use a

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1. What "Full Features" Requires Technically

For your phone to execute advanced hardware-level attacks, it must meet three conditions:

1. **An Unlockable Bootloader:** Allows you to flash custom partition images.
2. **Root Access (via Magisk or KernelSU):** Grants elevated administrative rights.
3. **A Custom NetHunter Kernel:** The standard Android kernel disables monitor mode and USB emulation. The custom kernel re-enables:

   Packet Injection & Monitor Mode (for Wi-Fi cracking).

   USB Gadget Drivers (for BadUSB / Rubber Ducky attacks).

   SDR (Software Defined Radio) Drivers (for listening to radio frequencies).

2. Best Phones to Buy for Full Features

Because building a kernel from source is complex, the easiest path to a full-featured NetHunter device is buying a phone that has **official pre-built kernel images** from OffSec:

Top Pick: **OnePlus 7 Pro / 8T (or OnePlus 6 / 6T)**

Why: OnePlus devices are widely considered the gold standard for NetHunter. Their bootloaders are trivial to unlock, and they have the most mature custom kernel support.

Feature Capability: HID attacks, BadUSB MITM, Bluetooth Arsenal, and custom OTG power delivery. Kali Linux

Budget Pick: **Xiaomi Poco F1 / Poco X3 Pro**

Why: Extremely cheap on the secondary market with huge community developer backing.

Feature Capability: Excellent full NetHunter kernel builds for Wi-Fi and USB gadget

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Classic Pick: **Google Pixel 3 XL or Pixel 4a**

> **Why:** Pixels have clean AOSP code bases, making flashing TWRP, Magisk, and NetHunter straightforward.

3. The Hardware Add-On (Essential for Full Wi-Fi Attacks)

Even on a fully rooted phone with a custom kernel, internal smartphone Wi-Fi chips often overheat or lack full 5GHz injection support. To unlock **100% wireless penetration capabilities**, you will need:

> **USB OTG Adapter / Cable:** Converts your phone's USB-C port to standard USB-A.
>
> **External Wi-Fi Adapter:** A USB Wi-Fi dongle with a chip that supports monitor mode (e.g., **Alfa AWUS036ACM**, **Panda PAU09**, or **TP-Link TL-WN722N v1**).
>
> Help Net Security

4. How to Get the Full Setup

```
[Unlock Bootloader] → [Flash Custom ROM / TWRP] → [Root with Magisk] → [Flash
```

1. **Buy a compatible device** (e.g., a used OnePlus 7 Pro or OnePlus 8T).
2. **Unlock the Bootloader** via Android Fastboot.
3. **Install TWRP Recovery** and root the device using **Magisk**.
4. **Download the Official NetHunter Installer** for your specific model from kali.org/get-kali/#kali-mobile.
5. **Flash the NetHunter** `.zip` in TWRP Recovery or via the Magisk Module Manager.

Where would you like to go next?

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Show exact setup steps for a OnePlus or Pixel device

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Recommend external Wi-Fi adapters for NetHunter

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Galaxy A06

If you are asking about using a **Samsung Galaxy A06** for Kali NetHunter, here is the direct breakdown: **it cannot support full NetHunter features (like BadUSB or Wi-Fi packet injection)**, but **it can easily run NetHunter Rootless** for software-based attacks.

The Reality of NetHunter on the Galaxy A06

❌ What Will NOT Work (Full NetHunter)

The Galaxy A06 runs a **MediaTek Helio G85** chipset. Scribd

- **No Custom NetHunter Kernel:** Offensive Security does not build official NetHunter custom kernels for budget MediaTek Samsung phones.
- **No Built-in Wi-Fi Injection or Monitor Mode:** The internal MediaTek Wi-Fi chip does not support monitor mode or frame injection.
- **No HID / BadUSB Attacks:** The stock Android kernel prevents USB emulation.
- **Knox Security:** Samsung Knox makes unlocking the bootloader and custom kernel flashing tricky, with risks of permanently breaking features like Samsung Pass.

✅ What WILL Work (NetHunter Rootless)

You can still run Kali Linux inside the phone using **Termux + PRoot**. This gives you a portable Linux environment to run software-level pentesting tools without bricking or rooting your phone:

- **Network Scans:** `nmap`, `masscan` (over standard TCP/UDP).
- **Web Security Testing:** `sqlmap`, `nikto`, DIRB, payload building.

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Full Desktop GUI: Running **NetHunter KeX** via VNC to get a graphical Kali desktop on your A06 screen.

Best Setup Strategy for Galaxy A06

If the Galaxy A06 is your current phone, here is the recommended way to use it for ethical hacking:

1. **Install NetHunter Rootless** via Termux (no root required) for terminal-based tools, scripts, and local network analysis.
2. **Use an External Wi-Fi Adapter (OTG):** If you ever root the phone and want wireless testing, buy an OTG adapter and an external USB Wi-Fi dongle (like an *Alfa AWUS036ACM* or *TP-Link TL-WN722N v1*) to handle packet injection over USB.

Scribd

How would you like to proceed with your Galaxy A06?

Guide: Install NetHunter Rootless on Galaxy A06

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Recommend cheap secondary phones for full NetHunter

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What is termux?

Termux is an Android terminal emulator and Linux environment app. It works directly without needing root access or special setup—giving you a minimal base system where you can install packages using the `pkg` or `apt` package managers.

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Think of it as turning your phone into a portable Linux command line.

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What Termux Allows You To Do

Run Linux Utilities: Access classic tools like `bash`, `zsh`, `nano`, `vim`, `git`, and `curl`.

Execute Programming Languages: Install and run Python, Node.js, Ruby, C/C++, Rust, and Go directly on your phone.

Security & Network Testing: Run networking and pentesting tools like `nmap`, `hydra`, and Metasploit, or use PRoot to run distributions like **Kali NetHunter Rootless**.

Manage Remote Servers: Use `ssh` to connect to remote Linux servers or cloud instances from anywhere.

How to Choose the Right File from Your Screenshot

Looking at the GitHub download screen in your image, here is which `.apk` you should download for your **Galaxy A06**:

👉 **Best Choice:** `termux-app_v0.118.3+github-debug_arm64-v8a.apk`

Why: The Galaxy A06 uses a 64-bit ARM processor (MediaTek Helio G85). This file is optimized specifically for your phone's processor and takes up much less space (33 MB).

👉 **Safe Alternative:** `termux-app_v0.118.3+github-debug_universal.apk`

Why: Contains code for all CPU architectures combined. It will work on any device, but the download is much larger (112 MB).

Note: *Avoid installing Termux from the Google Play Store, as that version is deprecated and no longer updated. Downloading the APK from GitHub (as shown in your screenshot) or from* **F-Droid** *is the correct method.*

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